The Socratic Method: Cross-Resolution Supervision for Papyrus Surface Segmentation

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Fig. 1. The intended visual argument. Given coarse CT and released M7, our selected cross-resolution student retains faint longitudinal sheet structure while removing cross-wrap blobs; the separately gated 2D fitter then repairs only short, geometrically supported gaps. The final registered crop will replace this reserved panel.

Coarse-resolution papyrus surface segmentation fails in two opposing ways: faint sheets disappear, while nearby wraps merge into blobs. We present a teacher-student method for Herculaneum papyrus data in which an approximately 2.399 μm native-fine model interrogates the released 9.362 μm M7 segmenter. The resulting student remains an ordinary, single M7 residual-encoder nnU-Net at inference. During training, projected soft occupancy is combined with medial-crest recall, teacher-background separation, M7 function preservation, a global parameter trust region, and dynamic widest-path connectivity. A separate additive 2D curve fitter reconnects short, geometrically credible gaps after segmentation. Our completed duration study improves both held-out scrolls over released M7 at every measured interval through 8,192 samples, with a best calibrated macro-scroll Dice of 0.58293 at 7,168 samples. Because its calibrated threshold lies at the lower boundary of the sweep, we select the operating point using independent morphology evidence instead: the raw 8,192-sample student at threshold 0.45. It passes the blinded anti-blob gate with no interior- or thickness-regression cubes, and an independent FLIP audit favors it over the 2,048-sample model on 12 of 16 locked slices. We release the recipe and selected candidate identity; weight redistribution remains subject to upstream terms.

CCS Concepts: • **Computing methodologies → Neural networks**.

Additional Key Words and Phrases: volumetric segmentation, knowledge distillation, papyrus, topology-aware learning, curve completion

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1 Introduction

The objects behind this work are the Herculaneum papyri, a library of scrolls carbonized by the AD 79 eruption of Mount Vesuvius. Carbonization preserved the rolls but left them too brittle to open and the writing trapped inside tightly wound coils. Synchrotron micro-CT can record their internal structure without touching the object; recovering a page then requires locating the papyrus sheet in three dimensions and *virtually unwrapping* it to a plane [Seales et al. 2016; Vesuvius Challenge 2026b]. A scroll is one continuous sheet, metres long, visiting many adjacent turns. Consequently, two pieces separated by only a few CT voxels may be distant along the page, while a short gap in the volume may interrupt one continuous line of text.

Segmenting the papyrus surface is the first geometric commitment in a scroll-unwrapping pipeline. A missed surface interrupts a sheet that may otherwise be traceable over a long distance. An overgrown prediction is at least as dangerous: it can weld nearby turns into one component and make a locally plausible surface globally wrong. A useful segmentation method must therefore recover faint structure and remove false connections at the same time. An overlap score alone does not adequately describe this tension.

The Vesuvius data provides a second, complementary source of information. A model operating on finer CT can resolve local evidence that is ambiguous after downsampling, while the deployed segmentation and geometry stack still consumes the coarser grid. We treat this mismatch as a cross-resolution dialogue. The fine model proposes soft occupancy and medial evidence; the released M7 model contributes a function that is already effective; explicit loss terms expose where those views are inconsistent. The student must reconcile them without moving far from M7.

This is the origin of our title. In the Socratic account, a speaker who professes little knowledge reveals inconsistencies in an expert's position and thereby points toward a better one. Our fine teacher is similarly not deployed as a superior replacement. It asks local questions of the expert coarse model. The answer is a revised M7 that stands alone at inference.

We present four contributions:

- a hash-pinned 2 μm-to-9 μm teacher-supervision recipe that can be executed independently of our geometry repository;

- an objective that balances soft occupancy, de-blob separation, medial recall, dynamic connectivity, and two forms of M7 preservation;
- a duration-ladder protocol that retains each declared observation rather than assuming the longest run is best; and
- an independently measured, additive 2D curve fitter that reconnects only short gaps able to answer geometric and cross-plane objections.

The 8,192-sample duration experiment and its locked morphology audit are complete. Although an overlap-only calibration prefers 7,168 samples at the censored lower threshold of 0.25, the independent anti-blob and human review select the raw 8,192-sample checkpoint at threshold 0.45. This disagreement is informative: the model is selected for retaining useful longitudinal structure without refilling nearby wraps, not for maximizing one overlap scalar.

2 Related Work

Segmentation. U-Net architectures and their self-configuring nnU-Net variants are widely used for dense volumetric segmentation [Isensee et al. 2021; Ronneberger et al. 2015]. Our deployed network follows the released M7 residual-encoder nnU-Net from the Vesuvius Challenge project exactly; we alter its supervision and optimization rather than adding an inference-time branch. This preserves compatibility with the existing coarse-volume pipeline.

Teacher supervision. Knowledge distillation transfers information from a teacher into a student by matching soft outputs or intermediate functions [Hinton et al. 2015]. Most such methods assume teacher and student predictions share a sampling grid. Our teacher observes a finer CT volume, so its occupancy and validity must be pulled back into the student's physical frame. The resulting targets represent partial volume rather than only a hard class.

Topology-aware Objectives. Skeleton and centerline losses preserve thin connected structures that can be under-represented by region overlap metrics [Shit et al. 2021]. The medial axis transform (MAT) instead describes a shape by centers and the radii of their maximally inscribed disks or spheres. LSMAT casts this construction as a least-squares problem to make the resulting representation less brittle under noisy or perturbed boundaries [Rebain et al. 2019]. This center–radius factorization is especially useful for papyrus: motion along the medial set extends a sheet, whereas increasing radius thickens it toward a neighboring wrap. We use that distinction to derive thin training corridors and M7 contact pins. Unlike a conventional skeleton overlap loss, our final connectivity term does not prescribe every voxel of one centerline. It raises the bottleneck of whichever admissible medial path the student currently supports. The atlas is constructed only from fully owned training events and never from held-out data.

Curve completion. Classical thinning extracts one-pixel skeletons from binary images [Zhang and Suen 1984]. Curve completion can then be posed as endpoint pairing under geometric continuity. For wound papyrus, however, a locally smooth completion can cross from one wrap to the next. Our postprocess therefore combines 2D tangents with an umbilicus-aware radial test, neighboring-plane corridors, explicit anti-merger and crossing gates, and connection persistence across slices.

Vesuvius models and data. Our experiments build on the public Vesuvius Challenge CT data, the released M7 surface model, and a Villa native-fine teacher [Vesuvius Challenge 2026a,b; Villa Model Authors 2026]. The final paper will replace these artifact-level entries with the authoritative model and dataset citations associated with the public release.

3 Overview

Fig. 2 summarizes our pipeline. We first run a native-fine teacher in its approximately 2.399 μm CT frame. Its probabilities, support, and medial crest are projected into the 9.362 μm grid as soft training fields (Sec. 4). We initialize every candidate from the exact released M7 checkpoint and train all parameters under a small global trust region. The output is one ordinary M7 network. At inference we use only its raw class-1 probability; there is no teacher and no M7 probability blend.

At the selected threshold, an optional C postprocess operates on each axial slice (Sec. 5). It proposes Bezier completions between skeleton endpoints, but paints a join only when local curve geometry, radial displacement, neighboring slices, and connection-track support agree. Segmentation training and curve completion are measured separately so an improvement in one stage cannot hide a failure in the other. All segmentation qualification numbers in Sec. 6 use the raw student without morphology or line fitting.

Two engineering details prevent accidental leakage or model substitution. First, the dynamic connectivity atlas is built only from boxes in the training manifest; held-out morphology is used only for evaluation. Second, the M7 initializer is accepted only if both its byte size and SHA-256 match the released artifact. A new candidate may resume its own interrupted optimizer state, but may not use an earlier student as learned initialization.

4 Cross-Resolution Supervision

4.1 Student Architecture and Data

Our student is a six-stage 3D residual-encoder nnU-Net with feature counts $(32, 64, 128, 256, 320, 320)$, one CT input channel, and two output classes. It has 102,349,770 trainable parameters. The frozen training schedule contains 4,096 deterministic 192^3 patches from PHerc0139 and a single native-fine teacher record. We traverse this schedule twice for 8,192 samples. The small corpus reflects the time available before this progress submission, not the extent of the registered data. Our broader pair plan contains three additional non-prize training scrolls—PHerc0332, PHerc1667, and PHercParis4—with 62,500 candidate origins allocated to each. None of the four training scrolls is eligible for a Herculaneum prize.

Our held-out validation manifest contains registered PHerc0814 and PHerc1451 patches. We evaluate every 1,024 samples and retain model-only checkpoints at 1,024, 2,048, 4,096, and 8,192 samples. Because the measured duration response is not monotonic (Sec. 6), we do not assume the final checkpoint is best.

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Fig. 2. Method overview. The final figure will separate training-only components (fine teacher, projection, auxiliary losses, and frozen M7 anchor) from deployed components (raw student and optional 2D fitter). It will also show that the parameter trust-region projection acts after every optimizer update.

4.2 Fine-Teacher Occupancy and Medial Geometry

The fine teacher provides two complementary targets in the student's physical frame. The first is soft occupancy $q \in [0, 1]$, obtained by an anti-aliased pullback from the 2.399 μm grid. It estimates how much of a 9.362 μm voxel is occupied by papyrus. The second is a binary medial crest c, projected with an OR/max operator so that the existence of a fine-scale center survives downsampling. Occupancy and crest have independent validity masks.

This separation follows a concrete failure of occupancy-only supervision. A thin but real continuation commonly occupies only 25–45% of a coarse voxel. Soft cross-entropy is correctly calibrated when it drives such a voxel toward $p = 0.25$–0.45, but the result may remain below the segmentation threshold. Reducing background separation moved both the center and its shoulders upward, recovering length while reintroducing thick blobs. We therefore retain q as an amount-of-material target and use c as an independent sheet-existence target.

LSMAT-style center–radius representation. LSMAT represents a shape using medial centers and their maximally inscribed radii rather than boundary samples alone [Rebain et al. 2019]. We use that center–radius view while retaining the released Villa teacher's center construction. For each native-fine axial slice T_z, the teacher skeletonizes the slice in 2D; the stacked centerlines receive one 3D morphological closing and are intersected with the original foreground. We pair every surviving center $u \in C_z$ with a physical 2D Euclidean-distance radius

$$r_z(u) = \min_{y \notin T_z} \|u - y\|_{\text{phys}}, \qquad \widehat{T}_z = \bigcup_{u \in C_z} B^\circ(u, r_z(u)). \tag{1}$$

The disks are open because the stored distance reaches the nearest background voxel center. Slice-wise radii are deliberate: pairing 2D centers with a 3D distance transform would mix disks and spheres and collapse the desired medial surface toward a curve. We audit reconstruction, component count, false-positive girth, and halo stability before using these centers for supervision.

The representation gives us two independent geometric coordinates. Moving along C_z extends the papyrus sheet tangentially; increasing r_z grows it radially toward another wrap. That distinction is the reason for introducing the LSMAT view: the connectivity loss should act along the first coordinate, while the separation loss and radius screens police the second.

From medial centers to pins. We construct events only on training slices with at least 95% valid teacher support. Components smaller than 12 voxels are removed. A slice qualifies when one thin teacher component H_e contacts at least two disconnected M7 components $M_{e,i}$ and contains at least two teacher voxels absent from the one-voxel M7 neighborhood. Both teacher and M7 must have maximum in-plane EDT radius at most three voxels, and no more than 2% of their foreground may survive a two-voxel erosion; these checks reject blobs before they can define a positive connectivity target.

Let C_e be the projected medial centers inside H_e, and let D_r denote radius-r dilation. We choose the smallest $d \in \{0, 1, 2, 3\}$ for which

$$\Omega_e = H_e \cap D_d(C_e), \qquad A_{e,i} = \Omega_e \cap D_2(M_{e,i}) \tag{2}$$

contains every nonempty contact set $A_{e,i}$ in one connected component. The sets $A_{e,i}$ are the *pins*: they are where distinct released-M7 fragments meet the LSMAT-style teacher-medial corridor, not arbitrary endpoints chosen from a student prediction. Corridor voxels within one voxel of M7, plus all pin voxels, form the free-anchor set F_e. Events touching their owning tile boundary are discarded, so every retained corridor is transformed and loaded as one unit.

4.3 Objective

Let p be the student's surface probability, q the projected fine-teacher occupancy, c its binary medial-crest field, $C = \{x : c_x = 1\}$ the corresponding valid crest set, and p_0 the released M7 probability. Our loss is

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{CE}}(p, q) + 0.25\mathcal{L}_{\text{Dice}}(p, q) + \mathcal{L}_{\text{crest}}(p, c) \\ &\quad + 2\mathcal{L}_{\text{sep}}(p, q) + 0.5\mathcal{L}_{\text{KL}}(p, p_0) + \mathcal{L}_{\text{pres}}(p, p_0) \\ &\quad + 0.03125\mathcal{L}_{\text{conn}}(p).\end{aligned} \tag{3}$$

All terms are evaluated only where their corresponding validity fields permit.

Soft Occupancy and Crest Recall. Our cross-entropy and Dice targets retain fractional occupancy created by fine-to-coarse projection. For a nonempty valid crest C, the additional term is

$$\mathcal{L}_{\text{crest}} = 1 - \frac{\sum_{x\in C} p_x + 1}{|C| + 1}. \tag{4}$$

We average Eq. 4 over crest-bearing samples rather than over all voxels. The normalization prevents sparse medial evidence from disappearing into the background count, and the additive one is the smoothing constant used by the released teacher loss. Because only crest voxels receive its gradient, the term asks for longitudinal sheet evidence without defining a thick target.

De-Blob Separation. Thick predictions need explicit shrink pressure. An M7 blob can overlap the fine sheet and still score well against an anti-aliased occupancy target. Let V and V_c denote occupancy and crest validity. We form the thin positive seed

$$P = C \cup \big(\{q \ge 0.5\} \cap (V \setminus V_c)\big), \tag{5}$$

using the crest wherever it is known and falling back voxelwise to hard occupancy only where it is not. We dilate P by two voxels and retain only occupancy-valid locations where the teacher is confidently background:

$$S = \big(\text{dilate}_2(P) \setminus P\big) \cap V \cap \{q \le 0.1\}. \tag{6}$$

The separation loss is the mean background cross-entropy

$$\mathcal{L}_{\text{sep}} = -\frac{1}{|S|}\sum_{x\in S}\log(1-p_x). \tag{7}$$

We weight this term by 2 in Eq. 3. Because S is a local fence around supported medial structure rather than arbitrary background, the term attacks thickness and short cross-wrap welds while the soft occupancy and crest terms remain free to grow a faint sheet elsewhere.

M7 function and positive preservation. The teacher is locally more informative, not globally complete. We therefore freeze a copy of released M7 during training and use its probability p_0 in two differently masked terms. Let $H = C \cup (\{q \ge 0.5\} \cap V)$, $h_x = \mathbf{1}[x \in H]$, and $m_x = \mathbf{1}[p_{0,x} \ge 0.5]$. The KL mask is

$$\begin{aligned} K = {} & \big((\overline{V}) \setminus D_2(H)\big) \\ & \cup \{x \in V : h_x = m_x \;\wedge\; (q_x \le 0.1 \;\vee\; h_x = 1)\}. \end{aligned} \tag{8}$$

Thus unknown space is anchored except for a radius-two corridor around known teacher positives. In known space, M7 is anchored only where its hard decision agrees with a confident teacher decision; ambiguous partial occupancy is not misclassified as agreement on background. We use

$$\mathcal{L}_{\text{KL}} = \frac{1}{|K|}\sum_{x\in K} D_{\text{KL}}\big(\text{Ber}(p_{0,x}) \,\big\|\, \text{Ber}(p_x)\big). \tag{9}$$

KL preservation alone can still allow a correct M7 foreground fragment to shrink under the separation term. We therefore define $R = V \cap D_2(H) \cap \{p_0 \ge 0.5\}$ and add the positive-only loss

$$\mathcal{L}_{\text{pres}} = -\frac{1}{|R|}\sum_{x\in R}\log p_x. \tag{10}$$

The two terms have separate roles: Eq. 9 retains M7's foreground and background function away from justified edits, while Eq. 10 prevents de-blobbing from deleting M7-supported sheet near the teacher. Teacher disagreement remains free to correct M7.

Dynamic medial connectivity. Separation supplies the desired shrink pressure, but without a complementary topological term it can also sever the same faint longitudinal connections we want to retain. Our first LSMAT axial loss connected the same pins with a minimum-off-axis spanning tree, then raised the weakest 10% of voxels on that fixed route toward probability 0.2. This proved that a thin, center-biased route could be supervised without using held-out geometry, but it also froze one of many equally valid paths. The lowest weight only tied the incumbent; larger weights monotonically reduced ordinary validation quality, and the strongest weight also reduced anti-blob passes. The final loss therefore retains the audited LSMAT corridor and pins but makes the path existential.

For event e, define the path capacity

$$a_e(x) = \begin{cases} 1, & x \in F_e, \\ p_x, & x \in \Omega_e \setminus F_e, \\ 0, & x \notin \Omega_e. \end{cases} \tag{11}$$

Taking $A_{e,0}$ as a source pin, initialize $r_e^0(x) = \mathbf{1}[x \in A_{e,0}]$ and iterate

$$r_e^{t+1}(x) = \mathbf{1}[x \in \Omega_e]\max\Big(r_e^t(x), \min\big(a_e(x), \max_{y\in\mathcal{N}_{26}(x)} r_e^t(y)\big)\Big). \tag{12}$$

This max–min recurrence computes the greatest bottleneck capacity among paths of at most t steps. After 96 steps, the event score and loss are

$$\begin{aligned} b_e &= \min_{i>0}\max_{x\in A_{e,i}} r_e^{96}(x), \\ \mathcal{L}_{\text{conn}} &= \frac{1}{|E|}\sum_{e\in E}\max\big(0, \text{logit}(0.2) - \text{logit}(b_e)\big). \end{aligned} \tag{13}$$

Each event receives one vote regardless of corridor size or pin count. Max/min back-propagation places pressure on the current best path's limiting capacity, not on every corridor voxel. Free M7 anchors have unit capacity and no direct gradient; no connectivity gradient exists outside Ω_e. The loss becomes exactly zero when every pin is reachable through some path with bottleneck at least 0.2.

The frozen atlas contains 149 fully owned events. Ninety-nine rows in the exact schedule encounter an event, 477 rows are eligible, and the longest audited event needs 44 of the allocated 96 propagation steps. Of these events, 135 have two pins, 13 have three, and one has four; 122 need no crest dilation and 27 need one voxel. The loss is evaluated only at the full-resolution output and has weight 0.03125. Together, the separation shell and existential path constraint express the two sides of de-blobbing: remove unsupported girth without deleting sheet existence.

Deep supervision. The occupancy, crest, separation, KL, and preservation terms are applied at the standard nnU-Net decoder outputs with normalized weights proportional to $1, 1/2, 1/4, \ldots$ and zero weight on the coarsest output. Soft occupancy is resized trilinearly. The sparse crest is max-pooled, while its validity is true only if every contributing fine cell is valid. Connectivity is intentionally excluded from lower-resolution outputs so resampling cannot broaden its medial corridor.

4.4 Optimization and Trust Region

We use Adam with constant learning rate 2×10^{-5}, betas $(0.9, 0.999)$, $\epsilon = 10^{-8}$, and zero weight decay. The batch size is three with eight-step gradient accumulation and bfloat16 autocast. Seed 1203 fixes the sample order and augmentation sequence.

After each optimizer update, we project the complete parameter vector θ into a relative-L^2 ball around released parameters θ_0:

$$\theta \leftarrow \theta_0 + \min\left(1, \frac{r\|\theta_0\|_2}{\|\theta-\theta_0\|_2}\right)(\theta-\theta_0), \qquad r = 0.0027535422. \quad (14)$$

The projection is active on essentially every observed update. It is thus part of the effective method rather than an inactive guardrail.

5 Additive 2D Curve Fitting

Following re-training, we introduce a downstream postprocess that repairs short omissions in the thresholded predictions. It is additive only: accepted joins set foreground pixels, but never erase the model output. This restriction makes every intervention easy to localize and prevents a gap-repair stage from quietly becoming a second segmentation method.

For each z-slice, we apply Zhang–Suen thinning [Zhang and Suen 1984], remove short spurs, label the 8-connected skeleton and mask components, and extract curve endpoints with outward tangents. We then snap endpoints to the mask edge so distances describe the visible gap rather than skeleton retraction. Endpoints clipped by the volume or missing data are retained for audit but may not be matched.

We score endpoint pairs using distance, facing, adjacent-plane evidence, and cross-slice track support. Before a pair enters deterministic greedy matching it must satisfy the following requirements:

(1) it does not close the same skeleton or mask component;
(2) its tangents face the gap and are sufficiently opposed;
(3) its reach is at most 12 pixels, and joins beyond six pixels have supporting evidence;
(4) if an umbilicus is available, its radial displacement is at most four pixels;
(5) its cubic Bezier has a bounded arc-to-chord ratio and does not approach a third component;
(6) the path corridor does not cross another sheet on nearby slices; and
(7) it neither crosses nor touches an already accepted join, and neither endpoint has taken a better match.

A first matching pass seeds endpoint and connection tracks. A second pass follows each new track through z, uses connection confidence, and retains only joins supported across at least three slices. We then paint a cubic Bezier disk stroke and record every added pixel.

6 Results

6.1 Evaluation Protocol

We calibrate thresholds from 0.25 through 0.60 in increments of 0.01 on registered PHerc0814 and PHerc1451 patches. Checkpoint ranking uses calibrated macro-scroll Dice with a minimum per-scroll gain guard against released M7. Final qualification evaluates the raw student at five durations (1,024, 2,048, 3,072, 4,096, and 8,192 samples) and thresholds 0.38–0.50, with 0.25 retained as a failure control. Its evidence is 16 locked PHerc0139 slices and six blinded PHerc1447 cubes. These locations are evaluation-only and spatially separated from the PHerc0139 training boxes. Promotion requires anti-blob behavior, two-sided morphology and coverage, and human review; no morphology, M7 blend, or line fitter is included in these model measurements. An earlier raw v29 candidate failed the PHerc1447 gate; we do not consider it a viable model.

Table 1. Completed v31 duration ladder. "Cal." is calibrated macro-scroll Dice; "t = .40" uses a fixed threshold; "gain" is calibrated macro gain over the M7 initial model. All calibrated thresholds equal the 0.25 search boundary.

Samples	Cal.	t = .40	Gain	Min. gain
1,024	.57944	.57136	+.01968	+.01917
2,048	.57886	.57086	+.01910	+.01868
3,072	.58037	.57253	+.02062	+.02037
4,096	.58022	.57196	+.02047	+.02024
5,120	.57873	.57014	+.01897	+.01813
6,144	.58104	.57329	+.02128	+.02124
7,168	**.58293**	**.57537**	**+.02317**	**+.02276**
8,192	.58037	.57226	+.02061	+.02031

6.2 Duration Study

Table 1 reports every committed validation interval through the completed 8,192-sample schedule. All intervals improve both held-out scrolls over released M7. The response is shallow and non-monotonic; the best is the 7,168-sample interval at 0.58293 calibrated macro-scroll Dice. This behavior justifies retaining the full ladder instead of exposing only the longest checkpoint.

The trust-region projection is active for 97.7% of optimizer updates in the first interval and 100% thereafter. The method therefore reaches its best observed validation score while remaining on the boundary of its allowed parameter neighborhood.

Every interval selects threshold 0.25, the lower edge of our sweep. The calibrated scores therefore compare duration, but do not determine the operating point: the apparent optimum is censored and 0.25 visibly refills the blobs that the independent gate is designed to catch.

6.3 Morphology-Gated Selection

Joint review selects the raw 8,192-sample checkpoint at $T = 0.45$. At $T = 0.42$ it obtains 16/16 exact component-count matches and already passes the PHerc1447 anti-blob gate. We nevertheless select 0.45 for its cleaner useful-structure/de-blob balance. At that operating point, 15/16 locked slices have exact component counts; the remaining case is a known scalar ambiguity in which the intended structures are present but one feature splits while two strokes touch.

On the six PHerc1447 cubes, the selected model has foreground ratio 0.950508 relative to the v15 anti-blob reference, reference recall 0.911324, zero interior-regression cubes, zero thickness-regression cubes, and a hard anti-blob pass. Human review finds useful line

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Fig. 3. Reserved cross-resolution supervision detail. The upper row will show a fine teacher mask, its LSMAT-style centers and radii, projected occupancy q, binary crest C, and the radius-two background shell S. The lower row will show a thin teacher component, disconnected M7 fragments, their contact pins, free anchors, the admissible medial corridor, and the dynamically selected widest path before and after its bottleneck is raised.

FIGURE RESERVED

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Fig. 4. Reserved line-fitter examples. We will show the thresholded input, traced endpoints and tangents, accepted and rejected proposals, connection support across z, painted pixels, and a registered high-resolution view. Rejections will include radial, third-component, cross-sheet, and crossing cases rather than only successful joins.

recovery in difficult views without restoration of the released-M7 blobs, although several undergrown views remain explicit failure watches. The full checkpoint hash is in the machine-readable release recipe.

Independent perceptual audit. We also run FLIP [Andersson et al. 2020] at 67.0206 pixels per degree on the exact 1,120 displayed candidate/teacher pairs. At $T = 0.45$, the 8,192-sample model improves FLIP on 12/16 locked slices relative to 2,048 samples; total teacher-only erosion falls from 4,283 to 4,270 pixels and false additions from 1,201 to 1,155. The audit is not unanimous: mean FLIP narrowly prefers 4,096 samples (0.232073 versus 0.233310), while weighted-median pooling narrowly prefers 8,192 (0.599964 versus 0.600288). We retain this disagreement. FLIP rewards teacher resemblance, whereas the blind gate and human review also penalize the teacher-like girth that can weld adjacent wraps.

6.4 Line-Fitter Evaluation

In a calibrated 4×5×5 study, the line fitter accepted 481 joins and added 12,697 pixels. Registered high-resolution precision was 92.5% overall and 98.4% for joins no longer than six pixels. We observed no full-turn fusion and measured a 21% reduction in atlas overlap. A larger 21-cube census measured 98.3% weighted precision.

Adjacent high-resolution connectivity is useful confirmation but is not, by itself, proof that a join remains on one wrap. The umbilicus-aware radial gate is our primary cross-wrap safeguard. The postprocessor evaluation therefore reports false joins by gate, distance, and radial regime rather than only an aggregate precision.

6.5 Interpretation

The duration experiment establishes consistent positive held-out gains under a saturated M7 trust constraint, while the completed morphology audit establishes a release candidate. Their different choices are the central result: overlap-only calibration prefers 7,168 samples and threshold 0.25, but the de-blob-aware evidence selects

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figure_2_registered_examples.pdf or figure_2_registered_examples.png

Fig. 5. Reserved registered comparison. Columns will show CT, released M7, projected fine-teacher supervision, the selected 8,192-sample output at $T = 0.45$, and two-sided add/remove errors. Rows will cover PHerc0814, PHerc1451, and the blinded PHerc1447 gate, with identical crops and thresholds clearly reported.

8,192 and 0.45. Table 1 is generated only from committed run artifacts, and the selection record preserves the independent locked, blind, and perceptual evidence rather than collapsing them into one score.

7 Limitations

Our fitting schedule is PHerc0139-only and may encode scroll-specific material or acquisition properties. Fine-teacher predictions are model-derived soft supervision rather than human ground truth. Two held-out calibration scrolls, 16 locked slices, and six blind cubes provide guards but do not establish broad robustness. The M7 trust projection is saturated, and its radius was selected under an earlier objective rather than the final loss in Eq. 3. Our overlap-calibration sweep also fails to bracket its optimum; threshold 0.45 is a morphology-gated operating choice, not the maximizer of that curve. Human review is necessarily less reducible than a scalar and should be repeated by independent readers before a public model claim.

The line fitter has a different failure mode. A completion may be locally smooth and persistent across slices but still join the wrong wrap near the core. Radial, third-component, and cross-sheet gates reduce this risk but cannot prove global correctness. The postprocess must remain independently measured and its additions must remain recoverable during qualification.

Finally, exact replay requires large, separately licensed artifacts. The training code, recipes, hashes, and export scaffolding are public-repository material, but the approximately 31 GB validation corpus, teacher atlases, M7 weights, and student checkpoints require dedicated artifact hosting and resolved upstream terms.

8 Conclusion

We presented a cross-resolution supervision method for improving the released M7 papyrus-surface segmenter without changing its deployed architecture. A fine teacher exposes local inconsistencies through soft occupancy, medial evidence, and de-blob separation; M7 preservation and a global trust region keep the answer close to the incumbent function. The same design principle extends downstream: the 2D fitter may propose a connection, but radial, merger, cross-sheet, crossing, and persistence gates force it to answer a sequence of geometric objections before adding pixels.

FIGURE RESERVED

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Fig. 6. Reserved failure cases: faint-sheet loss, residual blobs, cross-wrap risk, and examples whose morphology changes near the selected operating threshold.

All measured intervals improve both calibration scrolls. The best overlap diagnostic occurs at 7,168 samples, but locked and blinded morphology select the raw 8,192-sample checkpoint at $T = 0.45$. That choice passes the anti-blob gate with no interior or thickness regressions while retaining the longer run's longitudinal growth. The remaining release work is to package the selected weights under resolved upstream terms and replace the reserved panels with registered figures; the model decision itself is no longer provisional.

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